

APPENDIX F — VALIDATION, ANTI-SPOOF MEASURES, AND DATA INTEGRITY (NON-LIMITING)

A. Overview

This appendix provides non-limiting examples of validation methods, anti-spoof measures, and data integrity checks that may be used to improve robustness of measurements and reduce the likelihood of invalid sessions, simulated inputs, or misapplied devices. Any method described herein may be used alone or in combination.

B. Wear and Use-Context Validation (Non-Limiting)

In various embodiments, the system validates that a node is being worn or used in a valid context using one or more of: 1. contact integrity checks (pressure/contact baselines, capacitive contact confirmation, electrode impedance stability, optical coupling checks); 2. temperature dynamics indicating on-body coupling; 3. motion/orientation plausibility based on IMU signatures; and/or 4. multi-sensor agreement indicating coherent behavior across modalities. In some embodiments, the system requires a minimum validity duration prior to generating reportable metrics.

C. Session Validation for Geometry and Compliance Workflows (Non-Limiting)

In various embodiments, session-based workflows incorporate validation including: 1. scan-path validation (speed bounds, orientation bounds, smoothness bounds), 2. endpoint plausibility checks for intraluminal distance/geometry measurements, 3. stimulus-response plausibility for compliance workflows (monotonicity bounds, hysteresis bounds, pressure-response bounds), and 4. repeat-trial consistency requiring multiple passes to meet a repeatability threshold. In certain embodiments, the system outputs a session validity label and a scan confidence indicator.

D. Physiologic Confirmation and Anti-Spoof Indicators (Non-Limiting)

In various embodiments, anti-spoof measures include one or more of: 1. optical pulse confirmation (e.g., PPG-derived pulse features and pulse plausibility), 2. micro-motion signatures consistent with physiologic or on-body behavior, 3. thermal context plausibility indicating tissue-adjacent coupling, 4. impedance/dielectric plausibility indicating tissue-adjacent electrical properties, and/or 5. contact distribution plausibility indicating anatomically consistent coupling patterns without requiring explicit anatomical imagery. In some embodiments, the system requires a threshold number of confirmation signals (e.g., at least two modalities) before enabling cross-user correlation outputs.

E. Artifact Detection and Integrity Checks (Non-Limiting)

In various embodiments, artifact and integrity checks include: 1. saturation and clipping detection, 2. dropout detection and continuity scoring, 3. outlier detection relative to user baseline and/or population-referenced distributions, 4. environmental discontinuity detection (temperature jumps, wetness spikes), and 5. intra-node and cross-node consistency checks indicating whether correlated signals behave coherently. When anomalies are detected, the system may down-weight affected windows, label them as reduced confidence, exclude them, or request additional valid measurement windows.

F. Secure Pairing, Device Authenticity, and Tamper Resistance (Non-Limiting)

In various embodiments, data integrity and authenticity are supported by: 1. secure pairing between node and computing system, 2. authenticated message exchange, 3. cryptographic signatures or message authentication codes, 4. firmware integrity checks and update signing, and 5. device attestation mechanisms where supported. In certain embodiments, the system prevents unauthorized nodes from contributing data to correlation outputs.

G. Gating Rules for Cross-User Correlation (Non-Limiting)

In certain embodiments, cross-user correlation outputs are gated by one or more thresholds including: 1. minimum data recency, 2. minimum valid-window count, 3. minimum quality index thresholds, 4. minimum confirmation modality count, and/or 5. consent scope validation and active permission status. When gating thresholds are not met, the system may output a reduced-confidence indicator, a data insufficient state, or a recommendation for additional validated collection.

H. Variations

Any validation and anti-spoof method described herein may be applied to any node, any workflow, and any pairing context. Methods may be implemented on-node, on a user device, in a server environment, or in distributed configurations.